



Recognition and source memory for pictures in children and adults

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Abstract

The present experiment investigated the developmental aspects of source compared to item memory. College students and 7–8-year-old children viewed pictures drawn in red or green during a study phase, and were asked either to remember the pictures for a subsequent recognition test, or to remember both the pictures and their associated colors for a subsequent source memory test. In the test phase, new and old pictures were presented in black. In the recognition task, participants were asked to make binary old/new recognition judgments, while in the source task, they were asked to make trinary old-green/old-red/new source judgements. Performance on all tasks improved with increasing age, but the age difference for source was much larger than that for item memory. It has been suggested that the frontal lobes play a critical role in the retrieval of source information, and that this brain region relative to the medial temporal lobes continues to develop into late adolescence. Thus, it is possible that immaturity of the frontal lobes may be causally related to the children's lower performance on the source memory task. © 2001 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Within the episodic memory domain, it is possible to distinguish between memory about the occurrence of an event (i.e. content or fact memory such as recognition of a person's face), and memory for the context in which knowledge about the event was acquired (e.g. when and/or where one met the person). The latter aspect of episodic memory is referred to as source memory, whereas the former is referred to as item memory. Source memory can be based on temporal (when) or spatial (where) aspects of events, on the modality in which events were presented, on the perceptual details accompanying them, and on the emotional or cognitive state at the time the events occurred (for a review see [22]).

In everyday life, memory is typically better for facts than for their sources. While this phenomenon can be

partially explained by the amount of contextual detail available in the memory trace [7], recent studies have suggested that item and source memory are functionally and neuroanatomically dissociated. For example, healthy older adults, as opposed to young adults, have more difficulty recalling specific perceptual details accompanying facts, such as the modality of presentation [25,29], the letter case in which words were presented [23], and whether heard information was presented by a male or a female voice [24,40]. Similarly, older adults have more difficulty than younger adults in remembering the context in which they encounter events, such as whether they learned a fact during or before the experiment [19], whether they were instructed to state out loud or to think about a word [16], and which of two tasks they had performed when words were encoded [5].

Studies of amnesic patients have demonstrated that, among patients who show similar deficits in remembering facts, impairment of source memory may vary greatly [39,43]. These data further suggest that source memory may be dependent on brain structures other than the medial temporal lobes and diencephalon that

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are implicated in general amnesia. Because patients with frontal lobe lesions exhibited normal item recognition but disproportional numbers of errors regarding source information [19,32,33], it has been suggested that the frontal cortex is critical for accurate source memory. There is also behavioral and physiological evidence suggesting that the frontal lobes are particularly sensitive to the effects of aging [1,10,53], which is consistent with the decline in source memory that has been reported in the elderly. It is important to note that the contextual information (e.g. perceptual attributes such as color, as in the current investigation) encoded during an initial study episode is not likely to be stored in prefrontal cortex (e.g. [13]). Rather, this kind of information is probably stored in material-specific brain regions (where these perceptual attributes were originally processed) and is retrieved under the direction of a mechanism (or mechanisms) located in frontal cortical areas.

The frontal lobes also appear to have a long developmental course, as myelination [55] and synaptic density counts [17] do not appear to be complete until late adolescence. Because other brain regions that are assumed to be involved in memory functions such as the medial temporal lobes and the hippocampus are mature early in the course of development [34], a dissociation between item and source memory would be expected in children. Thus, the purpose of the present investigation is to address the issue of item versus source memory in children, as there is a paucity of information in this research domain (see below).

A developmental aspect of source memory has been addressed in the context of studies of reality monitoring, a paradigm which is a special case of source memory. In a reality monitoring task, the ability to discriminate memories of external events from those created internally (e.g. thoughts, imagining) is measured. Foley et al. [12] found that 6-year-old children had difficulty discriminating between memories of words they had actually said aloud and memories of words they imagined saying. However, they had no problem discriminating between words they imagined saying and words they heard spoken by someone else, or remembering which of two people had said a particular word. In another study, Foley and Johnson [11] concluded that young children are capable of distinguishing between memories for self generated actions and external events, but tend to be more confused than adults as to whether they performed or imagined the events (both were self generated events). The fact that, under some conditions, children failed to accurately monitor source was attributed to action similarity, person similarity, and to the involvement of the self (for more detail see [9,30]). While reality-monitoring studies showed a clear developmental trend, their focus differs in many respects from the neuropsychological approach

used to study source memory in the elderly and amnesic patients. For example, they lack a direct comparison with other forms of memory such as recognition, and their prediction is based on psychological theories [9] rather than on the developmental course of brain structures.

A neuropsychological approach was taken by Rybash and Colilla [38] who assessed 10–14-year-old children with the source-forgetting paradigm. In this task, the children were asked to discriminate between facts learned during the experiment and those learned at some other time in the child's life (this paradigm was based on Schacter et al. [39]). The children were also administered the Wisconsin card sorting test (WCST) and the verbal fluency task, which are presumed to assess frontal lobe functions. Rybash and Colilla [38] did not report the levels of performance for fact and source recall, nor performance on the WCST and verbal fluency tasks. Their major result was that error performance on the WCST was correlated with source recall error but not with item recall error, suggesting that children show poorer performance on source memory tasks requiring mature frontal cortex. However, the omission of the levels of item and source recall performance, and the lack of comparison with young adults, did not permit conclusions concerning the differential development of the item versus source aspects of episodic memory.

The purpose of the present investigation was to examine the development of the relationship between item and source memory, by comparing performance between children and adults. As there is some evidence that modality information is encoded automatically in both adults and children [26], while memory for colors appears to be an effortful process [15,35], color was used to define the context or source for initial learning during which children and adults were asked to memorize pictures and their colors for a subsequent memory test. Following the study phase, recognition memory tasks were administered in which participants were required to retrieve item as well as source information. It was expected that children would demonstrate relatively poorer source compared to item memory than adults. Two tests that are presumed to reflect frontal lobe function (i.e. the controlled word association test [27,52], and the competing programs task [31]), as well as a memory test that is assumed to depend on the medial temporal lobes (the story memory sub-tests [41,51]) were also administered to examine their correlation with the measures of item and source memory. If item and source memory are dissociable processes with distinct neurological underpinnings, then it was expected that performance on the frontal lobe tests would correlate with source but not item memory, whereas performance on the medial temporal lobe test would correlate with item but not source memory indices.

2. Methods

2.1. Subjects

Thirty-two children and 32 young adults, all native English speakers, participated in this study. The age and gender distribution within each group were: 7–9-year-old children (mean age, 8.05 ± 0.58 ; 14 female) and 18–24-year-old young adults (mean age, 19.85 ± 1.22 ; 16 female). Some of the children were recruited from a day school, and the remainder were referred by parents of these children. For their participation, children received a laboratory T-shirt. Young adults were undergraduate students who received course credit for their participation. Volunteers with a history of head trauma, neurological disorders, learning disabilities and hyperactivity were not included in the study. A socio-economic status index [50] was obtained based on educational level and occupation of the participants' parents, and did not differ between the groups. Informed consent was obtained from the young adults and from the parents of all children. Additionally, the 7–9-year-olds signed assent forms.

2.2. Neuropsychological tests

In order to assess participants' memory ability, story memory was administered to the adults from the Wechsler memory scale revised (WMS; [51]), and to the children from the wide range assessment of memory and learning (WRAML; [41]). These tests are assumed to assess memory functions that are subserved by the medial temporal lobe memory system [49,54]. There is evidence that, as in adults, medial temporal lobe structures are critical for storage and retrieval of events in children. The WRAML, similar to the WMS for adults, has been shown to be sensitive to medial temporal lobe damage in children [49,54].

Because there is evidence that source memory tasks involve activation of the frontal lobes (e.g. [19]), two additional tasks that are presumed to tap frontal lobe function were administered. Verbal fluency for the children and adults was measured with the controlled word association test [27,52], a task that shows a developmental progression [52], as well as strong relationships with other putative measures of frontal lobe function in children [28,52]. In this task, participants were asked to say, as quickly as they could, all the words they were able to think of beginning with a given letter (excluding proper nouns). A short practice was given with the letter 'C', and was followed by tests for the letters 'F', 'S', and 'A'. Participants were allowed one minute to provide the words for each letter, and their responses were recorded verbatim by the experimenter.

The competing programs task [31], which includes both visual and auditory components, was used as a second measure to assess frontal lobe function. In the visual portion, children and adults were instructed to tap twice with their index finger when the experimenter showed them one finger and to tap once when the experimenter showed them two fingers. Similarly, in the auditory portion when they heard two tapping sounds made by the experimenter they were asked to raise one finger and when they heard one tap to raise two fingers. Performance is measured by the number of correct responses, and by the number of correct switches (i.e. when the experimenter switches from tapping once to tapping twice).

2.3. Stimuli

The stimuli were 128 pictures (their modal names are presented in Appendix A) of common objects that were normed for children and adults [4,8,44]. Half of the pictures (64) represented animate concepts (i.e. four-footed animals, birds, sea creatures, and insects), and half represented inanimate concepts (i.e. clothing, furniture, kitchen utensils, toys, and vehicles). The 128 pictures were divided into four lists of 32 items each. Each list included 16 pictures from the animate and 16 pictures from the inanimate categories. Each of these lists was further divided into two sets, 'a' and 'b'. The lists were carefully constructed so that they were equated, separately for children and adults, on category membership, concept agreement, familiarity, and visual complexity (obtained from the norms provided in [4,8,44]). Statistical analysis of the variables characterizing the picture lists revealed no significant differences between the lists ($P_s > 0.1$). The lists were rotated across participants such that each list (and set) served equally often in all experimental conditions. An additional 32 pictures not appearing in the experimental lists were used for practice trials.

2.4. Procedure

The experiment was divided into four study-test blocks: two blocks for item recognition and two blocks for source memory. Young adults performed the four blocks consecutively, followed by the neuropsychological tasks, which took, on average, 45 minutes. The children were tested in two separate sessions, each including two blocks of the same test type (either item recognition or source memory), and each lasting about half an hour. The neuropsychological tests were administered to the children during the item recognition session. The order of the tasks was counterbalanced across subjects such that half the subjects were given the item recognition task in the first two blocks and the source memory task in the

last two blocks, while for half the subjects task order was reversed. In addition to the counterbalancing of the presentation lists, the pictures within a block were randomized separately for each participant. Participants were given a practice block before each task. All children (and occasionally adults) performed the practice block twice, to ensure that they understood the tasks.

Each block consisted of study and test phases. In the study phase, the participants viewed one set of 16 pictures from one of the lists. Half the participants saw set 'a' during the study phase and half saw set 'b'. In the test phase the participants viewed a total of 32 items, 16 of which they had seen in the study phase (test-old), and 16 of which were new items (test-new) from the other set of the same list. The same procedure was repeated with the other list of pictures for the rest of the blocks. Thus, each participant saw all 128 pictures, but as was mentioned, the order of presentation was completely balanced across participants. In order to enhance memory for the pictures, participants performed the study phase twice during which they engaged in a semantic decision task (see below).

2.4.1. Item recognition task

In the study phase, the participants viewed 16 pictures of which half were drawn in red and half were drawn in green. The participants were asked to make a semantic decision (hereafter referred to as the semantic discrimination task) by pressing one button on a response box if the picture represented an animate concept and a second button if the picture represented an inanimate concept. The animate and inanimate concepts were described to the children as living and non-living objects. In order to make sure that the children understood these categories, they were asked to give several examples of living and non-living objects. The hands assigned to 'animate' and 'inanimate' buttons were counterbalanced across participants. In addition, the participants were told to memorize the pictures for a subsequent memory test. Each picture was presented on the computer screen for 1500 ms, followed by a 1000-ms blank period, for a total ISI of 2500 ms. In the second presentation, the participants saw a different random ordering of the same pictures drawn in the same colors as for the first presentation.

In the test phase, the participants viewed a total of 32 pictures drawn in black. Of these, 16 pictures had been in the study phase drawn in either red or green (old), and 16 were new pictures (new). Participants viewed one picture at a time and were asked to press one button if they thought the picture was 'old', and a second button if they thought the picture was 'new.' Stimulus duration was 1500 ms with an inter-stimulus interval of 2500 ms. The hands assigned to 'old' and 'new' buttons were counterbalanced across participants.

2.4.2. Source memory task

The study phase (the semantic discrimination task) was identical to that of the item recognition task, with the exception that participants were also told to memorize the pictures' associated colors for a subsequent memory test. In the test phase, the participants viewed a total of 32 pictures drawn in black. Sixteen pictures had been seen in the study phase drawn in either red or green (old-red, old-green), and 16 were new pictures (new). Each participant viewed one picture at a time and was asked to press one of three buttons on the response box (two buttons with the thumb and index finger of one hand and one button with the index finger of the other hand). The participant was instructed to press one button if the picture had been seen before drawn in green (old-green), a second button if the picture had been seen drawn in red (old-red), and a third button with the other hand if the picture was 'new.'

2.5. Data analysis

The SPSS for Windows statistical package was used for all analyses. When a mixed design ANOVA was applied, group (adults, children) was the only between-subjects factor, and task (item, source) was one of the within-subjects factors. Other within-subjects factors will be specified when describing a particular analysis. Only significant ($P < 0.05$) main effects and interactions are reported. Effect sizes (f), computed according to Cohen [6] and Rosenthal [37], are reported for significant main and interaction effects. The convention provided by Cohen [6] was used with $f = 0.10, 0.25, 0.40$, indicating respectively small, medium, and large effect sizes. Post-hoc comparisons in the form of t -tests were used when significant findings had to be further interpreted.

3. Results

In general, adult performance exceeded that of the children on all tasks. In the following sections, the results for the study phase for both tasks are described first, followed by the results of the test phases, and then the comparison of item and source memory performance for both groups. These item and source indices will then be correlated with the neuropsychological measures in an attempt to demonstrate a double dissociation between item and source memory performance and tests presumed to tap, respectively, medial temporal and frontal lobe functions.

3.1. Semantic discrimination test (study phase)

As can be seen in Table 1, the overall mean percentage of correct discriminations in the semantic task is

very high for both adults and children. These data were subjected to a group by task by response type (animate, inanimate) mixed design ANOVA. For percent correct, there were main effects of task (97.5% for item recognition vs. 95.3% for source memory) ($F(1,62) = 26.95$, $P < 0.01$, $f = 0.44$) and group (99% for adults vs. 93.8% for children) ($F(1,62) = 60.61$, $P < 0.01$, $f = 0.98$), which were modulated by the task by group interaction ($F(1,62) = 6.95$, $P < 0.01$, $f = 0.11$). This interaction indicates that, while for both groups the additional memory demand during the study phase for the source memory task (remembering the color of the picture) reduced response accuracy as compared to the item recognition (old/new) task, this reduction in accuracy was larger for children (92.1 vs. 95.5%) than for adults (98.6 vs. 99.6%).

Fig. 1 presents the mean RTs for the first and second presentations of the pictures during the study phases of both the item recognition and source memory tasks. There are several findings that can be seen clearly in this figure. First, as was expected, the mean RTs of the children are longer than the mean RTs for the adults for both animate and inanimate pictures, and for both first and second presentations preceding both memory tasks. Second, for both groups, mean RTs during the study phase for the item recognition task are shorter than those for the source memory task. Third, for both groups, facilitation in RT can be seen for the second presentation relative to the first presentation of the pictures (repetition priming). Finally, only children show faster RTs for the animate than for the inanimate pictorial concepts.

These visual observations were supported statistically by a group by task by response type (animate, inanimate) by presentation (first, second) mixed design ANOVA of the RT data. As described above, there was a main effect of group ($F(1,62) = 21.29$, $P < 0.01$, $f = 0.34$), such that the young adults were faster (776 ms) than the children (924 ms). There was a main effect of task ($F(1,62) = 25.54$, $P < 0.01$, $f = 0.41$), such that study for the recognition test resulted in faster RTs (818 ms) than study for the source test (882 ms). In addition, there was a main effect of response type ($F(1,62) =$

30.28, $P < 0.01$, $f = 0.49$), where the RTs to inanimate objects were slower (859 ms) than those to animate ones (841 ms). There was also a main effect of presentation ($F(1,62) = 6.40$, $P < 0.05$, $f = 0.10$), such that second presentations were responded to faster (825 ms) than first presentations (875 ms). Group and response type interacted ($F(1,62) = 8.66$, $P < 0.01$, $f = 0.14$), indicating that only the children were slower in responding to the inanimate (943 ms) compared with the animate pictures (904 ms).

In summary, both groups demonstrated high performance in the semantic discrimination task, and both showed robust repetition priming. However, the difference in the encoding demands, i.e. memorizing pictures (for item recognition) vs. memorizing pictures and their associated colors (for source memory), resulted in lower response accuracy that was greater for the children than adults, and in longer reaction times for the source than for the item recognition task.

3.2. Test phase

3.2.1. Item recognition

Although old/new recognition was measured directly in the item recognition task, it can be computed indirectly for the source memory task. This can be done by collapsing the source memory data for correct old responses regardless of whether they were associated with a correct color (i.e. source) judgment. Table 2 presents the RT and accuracy data for both tasks. As expected, children were less accurate and responded more slowly than adults during both the item recognition and the source memory tasks. It is also clear from Table 2 that both groups were more accurate in recognizing old items as old during the source than during the item recognition task. A group by task ANOVA supported these visual observations. For probability of hits there were main effects of group (adults 97.1% vs. children 90.6%) ($F(1,62) = 27.27$, $P < 0.01$, $f = 0.44$) and task (92.5% for item recognition vs. 95.3% for source memory) ($F(1,62) = 9.48$, $P < 0.01$, $f = 0.15$) and no interaction. Similar results obtained when corrected hit rates (Pr)¹ were computed according to the methods described by Snodgrass and Corwin [39]. There were main effects of group (adults 0.93 vs. children 0.80) ($F(1,62) = 35.08$, $P < 0.01$, $f = 0.57$) and task (0.84 for item recognition vs. 0.88 for source memory) ($F(1,62) = 8.11$, $P < 0.01$, $f = 0.13$). Both

Table 1

Accuracy (%; mean $\pm$ S.D.) in the semantic discrimination task (study phase) preceding the item recognition and source memory tasks for adults and children

Group	Study Phase for	% Correct	
		Animate	Inanimate
Adults	Item	99.7 $\pm$ 0.92	99.4 $\pm$ 1.23
	Source	98.2 $\pm$ 2.94	98.6 $\pm$ 1.56
Children	Item	95.9 $\pm$ 4.15	95.2 $\pm$ 5.19
	Source	92.4 $\pm$ 5.06	91.8 $\pm$ 6.23

¹ Pr is the measure of discrimination for the two-high threshold model of recognition memory and represents the probability that either an old item will exceed the old item threshold or a new item will exceed the new item threshold. Br is the measure of bias for the two-high threshold model, and represents the probability of responding 'old' when in an uncertain state (i.e. when neither the old nor the new item recognition thresholds have been exceeded).

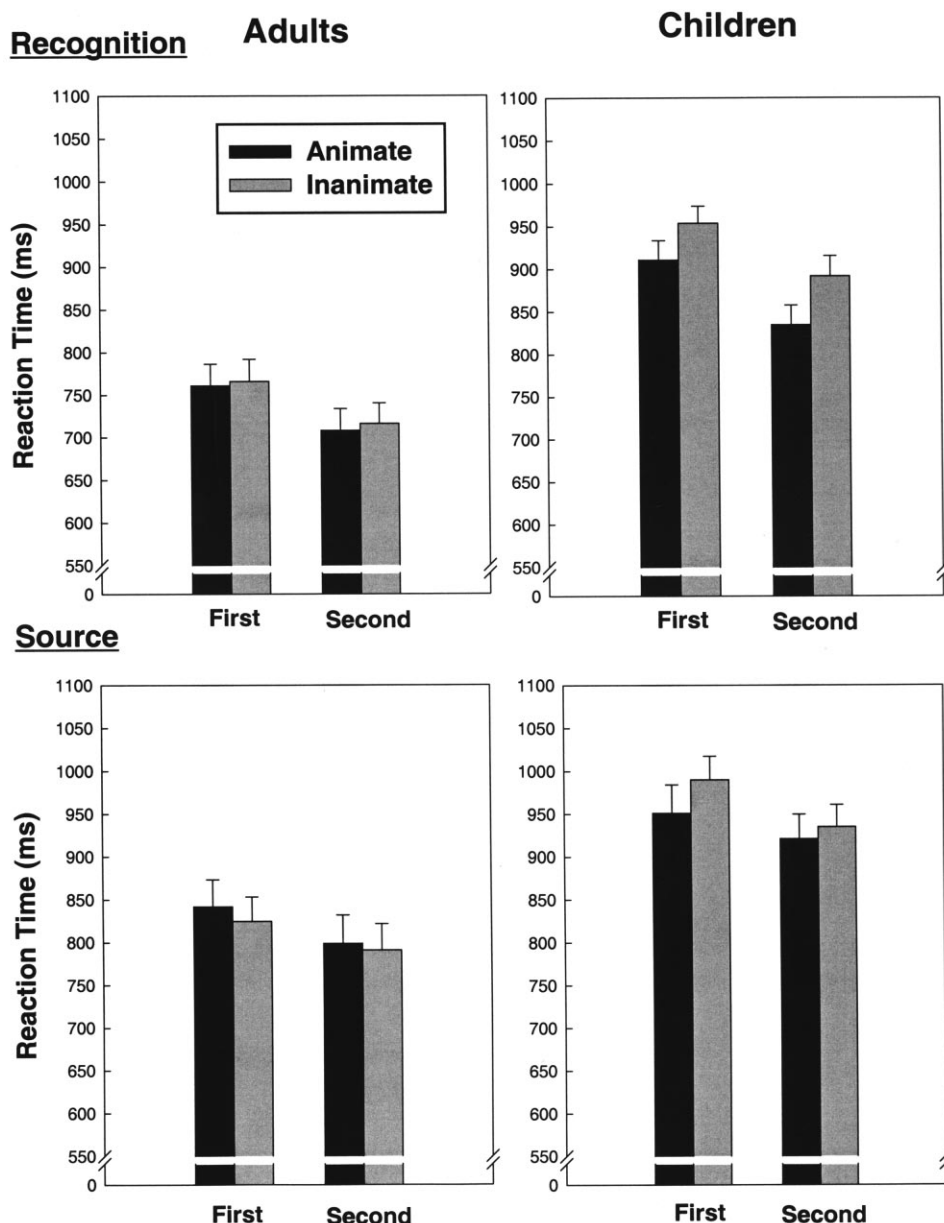


Fig. 1. Mean reaction times in the semantic discrimination task performed during the study phase of the item recognition and source memory tasks. The data are presented separately for first and second presentations. Vertical bars represent standard errors of the mean.

analyses indicate that adults performed better than children, and that performance was superior for the source than for the item recognition task. The bias measure, Br^1 (see [45]), did not vary across groups ($F < 1$) or tasks ($F < 1$) but was consistently below 0.5, indicating a conservative bias for both groups.

Reaction time also differed between groups and tasks. However, because the number of response buttons differed between the tasks (two-button press in the item recognition and three-button press in the source task) no direct comparison was made. However, it is clear that, while for the item recognition task participants responded faster to old than new pictures, the reverse was true for the source memory task. For each

task, the RT data were subjected to group by response type (old, new) mixed design ANOVAs. For the item recognition task, there were main effects of group ($F(1,62) = 77.07$, $P < 0.01$, $f = 1.24$), indicating that children were slower than adults (823 ms for adults and 1040 ms for children), and of response ($F(1,62) = 16.66$, $P < 0.01$, $f = 0.27$), indicating that the two groups produced faster RTs for old compared to new pictures (952 ms for new and 911 ms for old items). The interaction was not significant. For the source memory task, there was a main effect of group ($F(1,62) = 77.61$, $P < 0.01$, $f = 1.25$), again indicating that children were slower than adults (912 ms for adults and 1156 ms for children). The main effect of response type ($F(1,62) =$

Table 2
Mean RTs (ms) and S.D. for old and new pictures, probabilities of hits and false alarms (FA), discrimination (Pr) and bias (Br) for each group, and memory task^a

Task	Group	RT old	RT new	Hit	FA	Pr	Br
Item	Adults	809 (100)	837 (96)	0.95 (0.05)	0.03 (0.03)	0.91 (0.05)	0.43 (0.24)
	Children	1014 (112)	1067 (116)	0.88 (0.08)	0.11 (0.08)	0.77 (0.15)	0.46 (0.19)
Source	Adults	1036 (159)	787 (104)	0.97 (0.02)	0.03 (0.02)	0.94 (0.03)	0.45 (0.17)
	Children	1239 (151)	1072 (121)	0.91 (0.07)	0.09 (0.09)	0.82 (0.14)	0.48 (0.21)

^a Note: Pr and Br were computed as follows: $Pr = H - FA$; $Br = FA / (1 - Pr)$, where H refers to the hit rate, and FA to the false alarm rate.

126.25, $P < 0.01$, $f = 2.04$) revealed that for both groups old items elicited slower responses than new items (930 ms for new and 1138 ms for old items). However, group and response interacted ($F(1,62) = 4.89$, $P < 0.03$, $f = 0.08$) indicating that the increased RTs for old pictures were larger for adults ($\cong 250$ ms) than for children ($\cong 170$ ms).

3.2.2. Source memory

Source accuracy (I) is a commonly used measure in the source monitoring literature (e.g. [21,36]), and is an estimate of the conditional probability of a correct source identification given that the item was correctly recognized as old (see Table 3). While for both groups source accuracy was above the 0.5 chance level ($t_{adults}(31) = 23.87$, $P < 0.001$; $t_{children}(31) = 3.44$, $P < 0.001$), children were much less accurate than adults ($t(62) = 9.21$, $P < 0.001$). Inspection of the data also revealed that performance varied greatly among the children and less among the adults (see maximum and minimum values in Table 3). Therefore, the group of children was divided into two groups of high and low performers (16 subjects in each) based on their rank on the source accuracy measure, such that the high performers included the 16 highest scorers and the remaining 16 low scorers constituted the low performers. Table 3 presents the data of these two groups of children and shows that (a) source accuracy differed significantly between the two groups ($t(30) = 7.04$, $P < 0.001$); (b) source accuracy of the low performers was at chance ($t(15) = -1.50$; $P > 0.05$), whereas source accuracy of the high performers was above chance ($t(15) = 7.28$, $P < 0.01$), and (c) the high performers still performed significantly worse than adults ($t(46) = 5.59$, $P < 0.001$). The high and low performers did not differ on their item recognition scores (0.87 and 0.88, respectively), age, or socio-economic status. However, in all of the neuropsychological tests that were given to all subjects, the high performers showed a trend for better scores than the low performers (see Table 4).

Batchelder and Riefer [2] have pointed out several problems with the source accuracy measure (I). First, the measure combines performance across both sources (in the current case, over red and green colors) and thus does not distinguish between possible differential memorabil-

ity of the two sources. Second, the measure I is based on the hit rate alone without taking into consideration the false alarm rate, and thus may not fully capture the difference in source memory between the groups. Third, comparing source judgements across groups with unequal recognition accuracy could result in a confound. Thus, Batchelder and Riefer [2] offered an alternative means for analyzing the data using the multinomial model.

The multinomial model provides a way to estimate separately the overall detectability of old items regardless of their source (item recognition = D) from the discriminability of the items' sources (d) given that they were detected as old (conditional probability). Batchelder and Riefer proposed a series of restrictions on the model which can be tested against the data using the chi-square distributed fit statistic called G^2 . The restricted model which yields the lowest value of G^2 is selected as the preferred model. In the current data the model with the lowest value of G^2 (0.0057 for children and 0.0016 for adults) assumes that the probability of item recognition (D) and source memory (d) were identical for items presented in red and green. In other words, this model assumes that there was no difference in correctly detecting an old item as old when it was presented in either red or green, and there was no difference in correctly discriminating the item's color when it was presented in either red or green. This model further assumes that new items which are not classified as new are classified as old with probability ' b ', whereas the probability of guessing the source of an old item is ' a '. Finally, guessing the source of a non-detected (new) item is represented by ' g '. Table 5 presents the parameters produced by this multinomial model 5 (for more details see [2,3]).

The multinomial analysis demonstrates that the main difference between the groups is in source discriminability. As can be seen in Table 5, item recognition (D) for adults and children differs slightly but source memory (d) differs greatly. The latter finding is consistent with our analyses of source memory using the source accuracy measure I , demonstrating that the children are considerably different in their source discrimination than adults. In addition, children and adults did not differ dramatically in the three guessing parameters.

Table 3

Source accuracy (*I*) statistics for each age group and for high and low performers in the group of children^a

Group	<i>N</i>	Mean (<i>I</i>)	S.D. (<i>I</i>)	Min (<i>I</i>)	Max (<i>I</i>)
Adults	32	0.87	0.09	0.65	1.00
Children	32	0.59	0.15	0.33	0.94
High	16	0.71	0.11	0.57	0.94
Low	16	0.48	0.07	0.33	0.55

^a Note: S.D., standard deviation; Min, minimum score; Max, maximum score; High, high performer group; Low, low performer group.

As noted above, the young group performed at a lower level on both the item and source memory tasks. Therefore, a forced entry multiple regression approach was used to more precisely isolate and quantify the effect of age group on source task performance per se (i.e. to separate the contribution of poorer item memory on source memory performance). In this analysis, the dependent variable was the number of correctly assigned color items (source hits). The independent variables were entered in several steps. In the first step, the number of false alarms (to new items) was entered followed by the number of hits (to old items) from the item recognition task. In the last step, age group was entered. Thus, the effect of age on source memory performance was evaluated covarying for the effect of performance on item memory. The R^2 change for source memory hits due to the effect of age group, controlling for item performance was significant ($F(1,60) = 6.56$, $P < 0.013$), indicating that the children's performance on the source task differed significantly from that of the adults', above and beyond the children's attenuated performance on item memory. This analysis allows us to confidently conclude that the children's lower source memory performance is not due merely to their lowered (and correlated) performance on item recognition.

3.2.3. Item recognition versus source memory

Both item recognition and source accuracy were measured as probabilities. Therefore, it was possible to compare performance on the two tasks. Although the item recognition task was somewhat different than the source memory task, as can clearly be seen in Fig. 2, item memory (remembering the pictures themselves) appears to be very similar in both tasks. However, the slope for the source accuracy measure (remembering the colors of the pictures) appears to be much steeper than that of item recognition, suggesting that these two types of memory develop at different rates.

In summary, both item and source memory performance improved with age. However, this improvement was markedly greater for the source accuracy measure, despite the fact that children recognized the occurrence of old pictures with a high degree of accuracy.

3.3. Neuropsychological tests

As we predicted that adults would show better performance than children, independent sample one-tailed *t*-tests were used to compare the performance of adults and children on the three neuropsychological tasks. The raw scores, which are presented in Table 4, show higher scores for the adults than for the children on the verbal fluency measure and on three measures of the competing programs task (two from the auditory and one from the visual components), tasks which are assumed to tap frontal lobe functions. This was not the case for the story memory task, which is assumed to reflect medial temporal lobe function².

As mentioned in the Introduction, source memory and item recognition tasks are presumed to measure two different and potentially dissociable aspects of episodic memory that may be subserved by partially overlapping brain regions. Therefore, it was expected that performance on each of these tasks would be correlated with the neuropsychological task that is assumed to tap at least some of the functions of the associated neuronal tissue. That is, it was expected that performance on the source memory task would be correlated with performance on the verbal fluency and competing programs tasks but not with story memory. On the other hand, it was expected that item recogni-

² While the verbal fluency and the competing programs were administered in exactly the same format to children and adults, the story memory came from the WRAML for the children and the WMS-R for the adults (see Section 2). Choosing appropriate neuropsychological tests to compare among age groups is not simple. Using the same logical memory test for adults and children may result in poor performance by children not only because their memory is not mature yet, but also because the material is not age appropriate. Only by using a standardized test can we be sure that the test is appropriate for children and that it is reflecting similar cognitive functions as the test used with adults. The logical memory test taken from the WMS and WRAML are considered equivalent and a standard score (SS), which corrects for age, can be used to demonstrate that the two groups included individuals within the same normal range. For the purpose of our analysis, we did not use the SS but the raw scores, with the idea that this would enable us to compare performance in similar tasks. However, although unlikely, it is possible that the lack of a group difference may be an artifact of the different materials used to assess story memory by the two memory scales.

Table 4

Neuropsychological test scores (mean $\pm$ S.D.) for adults and children, and for the high and low performing subgroups of children^a

Task	Adults (<i>n</i> = 32)	Children (<i>n</i> = 32)	<i>t</i>	<i>P</i>	High (<i>n</i> = 16)	Low (<i>n</i> = 16)
<i>Story memory</i>						
Immediate	27.38 $\pm$ 5.17	27.22 $\pm$ 6.28	0.11	ns	28.18 $\pm$ 5.71	26.25 $\pm$ 6.84
Delayed	24.31 $\pm$ 5.69	24.78 $\pm$ 6.59	−0.31	ns	25.38 $\pm$ 6.77	24.19 $\pm$ 6.57
Verbal fluency	47.56 $\pm$ 10.97	28.81 $\pm$ 10.27	7.06	0.001	31.94 $\pm$ 12.02	25.69 $\pm$ 7.26
<i>Competing programs</i>						
Auditory	18.47 $\pm$ 1.08	16.53 $\pm$ 2.62	3.86	0.001	17.00 $\pm$ 2.97	16.06 $\pm$ 2.24
Auditory switch	7.06 $\pm$ 1.05	6.31 $\pm$ 1.60	2.23	0.03	6.69 $\pm$ 1.44	5.94 $\pm$ 1.69
Visual	19.53 $\pm$ 0.76	19.09 $\pm$ 1.23	1.71	ns	19.37 $\pm$ 0.96	18.81 $\pm$ 1.42
Visual switch	7.78 $\pm$ 0.49	7.47 $\pm$ 0.72	2.03	0.05	7.62 $\pm$ 0.62	7.31 $\pm$ 0.79

^a Note: story memory from the WRMAL for the children and WMS-R for the adults. High, high performing group of children; Low, low performing group of children; ns, not significant.

tion performance would be correlated with scores on the story memory task, but not on putative tests of frontal lobe functions. To investigate this possibility, the following variables were entered into the correlation matrix: hit rates from the item recognition and source memory tasks, source accuracy (*I*), and the neuropsychological test scores presented in Table 4. Partial correlations controlling for age were performed, and one-tailed tests of significance were used because of the predicted direction of the correlational pattern. Table 6 presents the matrix of the partial correlations with age related variance removed. As was expected, the measures of item recognition (both measures of hit rate) were significantly correlated with the two-story memory scores (immediate and delayed story recall) but not with the measures of frontal lobe function. On the other hand, the source accuracy measure was significantly correlated only with the two measures of the competing program task (number of correct responses and correct switches for the visual task), but not with the story recall measures. Thus, these data demonstrate a double dissociation between item and source memory performance and tests that tap the putative brain regions on which they are thought to depend.

4. Discussion

The present study demonstrated that performance on fact (item recognition) and source memory improves with age. However, it appears that children had relatively more difficulty in retrieving source information as compared to retrieving item information only. Thus, these data provide new evidence, based on developmental data, to support the notion that source and item memory reflect the activation of functionally (and probably neuroanatomically) separate systems. The present data also extend the findings of developmental trends reported in reality monitoring experiments [11,12,30] to include a type of source memory that does not involve the self or specific actions.

The longer reaction times and reduced accuracy for each of the groups during the encoding phase of the source memory as compared to the item recognition task indicate that encoding color information required more mental resources than memorizing the pictures alone. It is possible that the additional attention devoted to encoding the picture's color facilitated the binding of item memory and context [20] and this resulted in improved performance in old/new recognition during the source memory compared with the item recognition task for both adults and children. However, remembering the colors of the pictures was disproportionately lower in children compared to adults, and appears to be dissociated from the children's item memory. In particular, this was demonstrated in the analysis of the high and low performers within the childhood sample. In that analysis, children who scored high on the source memory task had the same mean score on the item recognition task as did children who performed at chance in the source memory task.

The fact that reaction time to old pictures was faster than that for new pictures during the item recognition test but was slower for old than for new pictures during the source memory test may be explained by the differ-

Table 5

Estimated item and source memory probabilities from application of the one-high threshold model^a

Parameters	Adults	Children
<i>D</i>	0.98	0.92
<i>d</i>	0.74	0.18
<i>a</i>	0.42	0.50
<i>g</i>	0.60	0.57
<i>b</i>	0.01	0.07

^a Note: *D*, probability of detecting old item as old; *d*, probability of detecting red item as red or green item as green; *a*, probability of guessing the source of a detected red or green item; *g*, probability of guessing a non-detected green or red item; *b*, probability of guessing a new item as old.

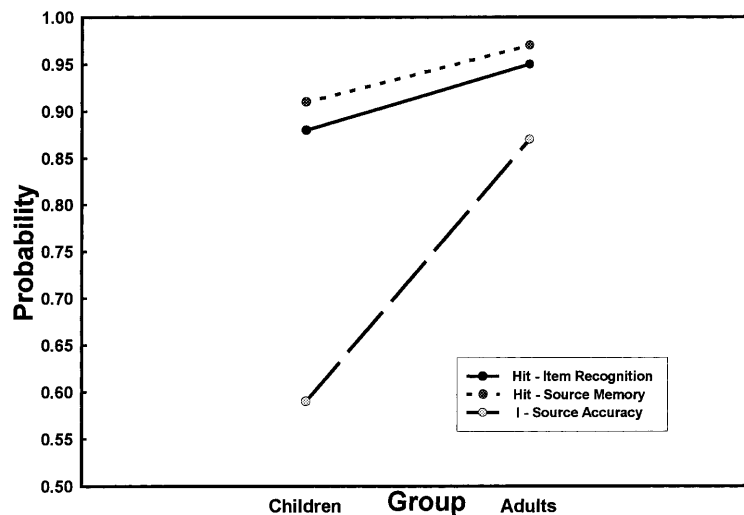


Fig. 2. Probabilities of correctly detecting old pictures (hits) in the item recognition and the source memory tasks for both adults and children. Source accuracy scores for the source memory task are also presented for both age groups.

ent number of button presses that were used in each test. It is also possible that this difference reflects the greater difficulty in performing the source memory test. Moreover, the greater success, during the source memory test on behalf of the adults, together with their larger increase in reaction time for old relative to new pictures (as compared with children), may indicate that elaborative strategies are needed to successfully perform the task. It is, thus, possible that such strategies may not have been available to the children who showed a smaller increase in RT for old relative to new pictures, as well as poorer source memory.

Strategic behavior is one of the many functions ascribed to the frontal lobes. Other functions, such as integration of information from multiple brain regions and attentional control, are also under the direction of the prefrontal cortex. The frontal lobes appear to mature slowly during childhood relative to other brain regions. Thus, it is possible that the relatively lower performance of the children during the source memory task stemmed from the fact that they were not yet capable of integrating conceptual and perceptual information necessary for accurately retrieving source (color) information.

This dissociation between fact and source memory has been reported for patients who suffer from various types of memory disorders. For example, memory for temporal order information is worse in amnesic patients with Korsakff's syndrome (who often show frontal lobe pathology) relative to amnesic patients with medial temporal lobe pathology [42,48]. These two groups of amnesic patients did not differ, however, in their item memory scores, both of which were impaired to the same extent.

The deficit in temporal order memory has also been observed in patients with frontal lobe pathology, and

was correlated with impairment on neuropsychological tests sensitive to frontal lobe damage [47]. While the children in the present study did not suffer from any known neurological disorder, they did show inferior performance relative to the adults on the neuropsychological tests presumed to reflect frontal lobe function. Moreover, in the correlational analysis, there was a double dissociation between item and source memory performance and performance on tests presumed to tap the functions of the brain systems on which they are thought to depend. Item recognition was correlated with neuropsychological test performance presumed sensitive to medial temporal lobe function, but not with test performance presumed to tap frontal lobe function. In contrast, source memory accuracy scores were correlated with neuropsychological test performance presumed sensitive to operations of the frontal lobes, but not with test performance presumed to assess functions of the medial temporal lobes. Thus, the present study with children and adult subjects supports a similar double dissociation as that reported by Glisky et al. [14] who tested two groups of elderly participants. Furthermore, these data, together with previous evidence that the maturational rate of frontal brain regions lags behind that of hippocampal regions [17,34,55], supports our initial hypothesis that children's ability to remember item information would be dissociated from their ability to remember contextual (or source) information.

While the data demonstrated that source memory is a dissociable component of episodic memory, and develops relatively slowly, it is not yet known at what age the ability to remember source information accurately is completely mature. The extent to which this difficulty in remembering the pictures' colors can be generalized to other types of contextual information such as temporal order and spatial location, is also not known. In

Table 6

Matrix of partial correlations controlling for age between neuropsychological test scores and item and source scores^a

	MemI	MemD	VF	AudC	AudS	VisC	VisS	HitR	HitS	<i>I</i>
MemI	–	0.88*	–0.08	0.18	0.18	0.09	–0.05	0.31*	0.25*	<i>0.13</i>
MemD		–	0.07	0.14	0.12	0.05	–0.14	0.31*	0.28*	<i>0.06</i>
VF			–	0.08	0.03	0.05	–0.03	–0.03	0.14	0.18
AudC				–	0.81*	0.15	0.11	0.14	–0.13	0.11
AudS					–	0.07	0.04	0.16	–0.01	0.12
VisC						–	0.81*	<i>0.10</i>	–0.01	0.39*
VisS							–	<i>0.10</i>	–0.01	0.31*
HitR								–	0.31*	0.03
HitS									–	0.29*
<i>I</i>										–

^a Note: scores for story memory are: MemD, delayed recall; MemI, immediate recall. Scores for competing programs are: AudC, auditory correct; AudS, auditory switch; VisC, visual correct; VisS, visual switch; VF, verbal fluency; HitR, item recognition during the old/new task, HitS, item recognition during the source task; *I*, source accuracy. Italic and bold fonts indicate expected direction of correlational effects.

* Significance at the 0.01 level.

addition, precisely which processes can be subserved under ‘frontal lobe immaturity’ is not yet known. It may be that the lower source memory performance found in children is due to differences in attentional capacity and allocation [18], and/or to limited intercorrelational interactions in the immature brain [46] that are necessary for adequate performance during complex memory tasks. Future studies with children of different ages and with different contextual sources should enhance our understanding not only of the developmental aspects of distinct memory functions, but also the nature of those subcomponents that make up a multifaceted memorial representation.

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Appendix A. The modal names of the pictures used in the present study

A.1. List 1

Animated: bee, caterpillar, dolphin, fox, giraffe, gorilla, kangaroo, lamb, lion, octopus, panda, parrot, pig, turkey, turtle, vulture.

In-animated: ball, baseball bat, bowl, clock, coat, desk, dress, fork, kettle, ladle, motorcycle, rolling pin, sailboat, shoe, train, vase.

A.2. List 2

Animated: bear, buffalo, butterfly, elephant, horse, ladybug, lizard, owl, rhinoceros, scorpion, seal, skunk, swan, swordfish, tiger, toucan.

In-animated: airplane, bus, chair, chest, couch, cup, garbage can, glass, hat, knife, roller skate, swing, television, vest, wagon, yo-yo.

A.3. List 3

Animated: alligator, camel, cat, crab, cow, dog, flamingo, grasshopper, koala, monkey, ostrich, rabbit, shark, sheep, snail.

In-animated: baby carriage, bed, bench, blouse, bottle, frying pan, helicopter, kite, pants, record player, refrigerator, sock, stove, toaster, top, tractor.

A.4. List 4

Animated: beetle, chicken, deer, dinosaur, duck, eagle, frog, hippopotamus, leopard, mouse, seahorse, snake, spider, squirrel, starfish, zebra.

In-animated: balloon, bicycle, car, lamp, mitten, picture, pitcher, pot, salt shaker, sled, spoon, stool, table, tie, truck, whistle.

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